

2.1 Mathematical Background

Consider a two-layer neural network

$$h(x; \theta) = \frac{1}{\alpha(m)} \sum_{i=1}^m a_i \sigma(w_i^\top x), \quad (3)$$

where $x \in \mathbb{R}^d$, $w_i \in \mathbb{R}^d$ are the inner weights, $a_i \in \mathbb{R}$ are the outer weights, $\sigma(\cdot)$ is the activation function, and $\alpha(m)$ is a scaling factor defining the regime. At initialization, $W_{ij} \sim \mathcal{N}(0, 1)$, with $\mathbb{E}[a_i] = 0$ and $a_i \perp W_{ij}$, so that $\mathbb{E}[h(x; \theta)] = 0$ and $\text{Var}(w_i^\top x) = 1$. For n inputs $X \in \mathbb{R}^{n \times d}$ with $X \sim \mathcal{N}(0, I_d/d)$, (3) generalizes naturally to matrix form where $W \in \mathbb{R}^{m \times d}$. The three regimes differ only in $\alpha(m)$, and are designed to keep $\text{Var}(h(x))$ at a controlled scale as $m \rightarrow \infty$.

2.1.1 Mean Field (MF)

Setting $\alpha(m) = m$ gives $\text{Var}(h_{\text{MF}}(x)) = \frac{1}{m} \text{Var}(a_i)$ and $\text{Var}(\sigma(w_i^\top x)) = \frac{1}{m} \cdot 1 \cdot 1 = O(\frac{1}{m})$, so the variance of the network shrinks to zero as $m \rightarrow \infty$. Therefore by the law of large numbers, $\lim_{m \rightarrow \infty} h_{\text{MF}}(x) = \mathbb{E}[\Psi]$, where $\Psi(a, w): x \mapsto a\sigma(x^\top w)$ [1, 6].

2.1.2 Neural Tangent Kernel (NTK)

Setting $\alpha(m) = \sqrt{m}$ [2] keeps the variance stable:

$$\text{Var}(h_{\text{NTK}}(x)) = m \cdot \frac{1}{m} \text{Var}(a_i) \text{Var}(\sigma(w_i^\top x)) = O(1),$$

so unlike MF, the output variance doesn't vanish with width. Both inner and outer weights are trained.

2.1.3 Random Features (RF)

The RF regime [8] uses the same scaling $\alpha(m) = \sqrt{m}$, and therefore has the same $O(1)$ output variance as NTK. The key distinction is that only the outer weights a_i are trained; the inner weights w_i are fixed at initialization. This reduces (3) to a linear model with random features $\Phi_i(x) = \sigma(w_i^\top x)$.

2.2 Experiment 1: Performance Comparison Across Regimes

2.2.1 Objective

This experiment compares training and generalization performance across all three regimes (MF, NTK, RF) to understand which regime converges fastest, which activation function performs best, and how performance scales with network width m . We expect Mean Field to perform the best as its learning rate scales with width and Random Features to perform the worst as it learns less features.

2.2.2 Experimental Setup

The network (3) was evaluated on three datasets: a synthetic dataset, MNIST [4], and CIFAR-10 [3].

Synthetic Dataset Labels were generated as $y_j = h^*(x_j)$, where h^* is a ground-truth network of the same form as (3) with fixed parameters $\{a_i^*, w_i^*\}$ and $\sigma^* = \text{ReLU}$. Inputs were drawn as $x_j \sim \mathcal{N}(0, I_d/d)$, ensuring $\mathbb{E}[\|x_j\|_2^2] = 1$ across all dimensions.

CIFAR-10 and MNIST Pre-processing Both datasets were reduced to binary classification (classes 0 & 1) and images were flattened to $d = 784$ (MNIST) and $d = 3072$ (CIFAR-10). Each sample was then standardized as $x_j \leftarrow \frac{x_j - \bar{\mu}}{\bar{\sigma}\sqrt{d}}$, so that $\mathbb{E}[\|x_j\|_2^2] \approx 1$, matching the $\mathcal{N}(0, I_d/d)$ assumption of (3).

Evaluation Framework All regimes were trained across $\sigma \in \{\text{ReLU}, \text{erf}, \text{tanh}\}$, widths $m \in \{100, 200, 400, 800\}$, and $T = 1000$ iterations. Synthetic experiments used $d = 10$, $n = 3000$, $n_{\text{test}} = 500$, $m^* = 20$; real datasets used $n = 4000$, $n_{\text{test}} = 1000$. Learning rates $\beta_{\text{MF}} \in \{0.03, 0.1, 0.5\}$ and $\beta_{\text{NTK}}, \beta_{\text{RF}} \in \{0.1, 0.5, 1.0\}$ were tested. According to Section 2.3, $\beta_{\text{MF}} = 0.1, \beta_{\text{NTK}} = \beta_{\text{RF}} = 1.0$ are the optimal parameters for each regime's learning rate hence why they are included. The other choices of β are tested for comparison. All experiments minimized the mean squared error (MSE) loss and were run with seed = 42.

2.2.3 Results and Analysis

Results are shown in Table 1. Full synthetic dataset results are presented in Table 3. Full MNIST results are shown in Table 4 and CIFAR-10 results, in Table 5.

Table 1: Best regime configurations by test loss, grouped by dataset and regime with β in ascending order. Generalization error is computed as (test – train). For each dataset, **bold** marks the lowest value in each numeric column and underline marks the highest.

Dataset	Regime	β	σ	m	Train Loss ↓	Test Loss ↓	Gen. Error ↓
Synthetic	MF	0.03	ReLU	200	0.0838	0.0805	−0.0033
	MF	0.1	ReLU	100	0.0236	0.0253	+0.0017
	MF	0.5	ReLU	400	0.0021	0.0034	+0.0013
	NTK	0.1	ReLU	100	0.1627	0.1717	+0.0090
	NTK	0.5	ReLU	200	0.0543	0.0555	+0.0013
	NTK	1	ReLU	100	0.0320	0.0393	+0.0073
	RF	0.1	ReLU	100	<u>0.4966</u>	<u>0.5155</u>	<u>+0.0189</u>
	RF	0.5	ReLU	800	0.1367	0.1435	+0.0068
	RF	1	ReLU	800	0.0842	0.0796	− 0.0046
MNIST	MF	0.03	ReLU	200	0.0094	0.0343	+0.0249
	MF	0.1	ReLU	200	0.0048	0.0238	+0.0190
	MF	0.5	erf	800	0.0027	0.0102	+ 0.0075
	NTK	0.1	ReLU	200	0.0786	0.1391	+0.0605
	NTK	0.5	ReLU	100	0.0225	0.0897	+0.0672
	NTK	1	erf	400	0.0380	0.0589	+0.0209
	RF	0.1	ReLU	100	<u>0.1074</u>	<u>0.1842</u>	<u>+0.0768</u>
	RF	0.5	ReLU	400	0.0440	0.1077	+0.0637
	RF	1	ReLU	400	0.0286	0.0784	+0.0498
CIFAR-10	MF	0.03	ReLU	800	0.1342	0.1269	−0.0073
	MF	0.1	ReLU	800	0.1051	0.1088	+0.0037
	MF	0.5	ReLU	800	0.0415	0.0946	<u>+0.0531</u>
	NTK	0.1	ReLU	100	0.1886	0.1843	−0.0043
	NTK	0.5	ReLU	100	0.1365	0.1392	+0.0027
	NTK	1	ReLU	100	0.1236	0.1297	+0.0061
	RF	0.1	ReLU	100	<u>0.2115</u>	<u>0.2041</u>	− 0.0074
	RF	0.5	ReLU	100	0.1538	0.1530	−0.0009
	RF	1	ReLU	200	0.1399	0.1417	+0.0019

Activation and Width ReLU is the dominant activation, achieving the lowest test loss in 25 of 27 configurations in Table 1; both exceptions occur on MNIST, where erf beats ReLU for MF at $\beta=0.5$ (0.0102 vs. 0.0196) and for NTK at $\beta_{\text{NTK}}=1$ (0.0589 vs. 0.0760). This should be no surprise for the synthetic experiments as the ground truth activation function used was ReLU. Optimal width is regime-dependent rather than dataset-dependent: MF consistently favors $m=800$ on MNIST and CIFAR-10, while NTK prefers $m=100$ on essentially every setting. This split follows from the theory; MF’s effective step size $\eta=\beta \cdot m$ makes width directly useful, whereas NTK’s variance is stable despite changes in m .

Learning Rate No single β is universally best across regimes though this is purely based on performance metrics (Section 2.3 offers a better explanation as to which choice of β is best). MF benefits monotonically from larger β on every dataset (e.g., synthetic MF goes $0.0805 \rightarrow 0.0253 \rightarrow 0.0034$ as β rises from 0.03 to 0.5). Furthermore, it can be seen that the best MF configurations have a high m (800 in MNIST and CIFAR-10, 400 in synthetic). This corresponds to the scaling of the optimal learning rate of MF $\eta_{MF} = \beta \cdot m$. NTK and RF, whose per-step updates lack the